

Detailed Design, Data Models, and Interface Contracts

COMP-7431: Advanced Software Engineering | Week 7
CampusConnect: adding a dependable voice path

Model it. Contract it. Mock it. Connect it. Test it.

Week 7 Resumes at the Architecture-to-Design Handoff

AFTER THE HOLIDAY + MIDTERM

Week 4 selected an architecture. Week 7 must make every boundary precise enough to build, mock, and test.

THE QUESTIONS DETAILED DESIGN ANSWERS

- Which component owns each decision?
- What data crosses each boundary?
- What does success look like?
- What errors can occur?
- Where do secrets live?
- What happens when voice or AI fails?

Architecture chooses the parts. Detailed design turns their relationships into enforceable agreements.

CampusConnect Needs Voice—Without Losing Trust

NEW EVIDENCE FROM END USERS

Students want hands-free IT help. Some also need an alternative to typing or reading a dense screen.

THE DESIGN QUESTIONS

- What exactly does the agent say first?
- Which documents may ground an answer?
- How will it speak and display a source?
- When must it abstain?
- How long may each turn take?
- How can the same journey continue in text?

The feature request is “add voice.” The design task is to preserve truth, access, and recovery across channels.

A Contract Is an Observable Promise Between Components

“If two components can disagree about a message, state, error, or timing rule, the design is unfinished.”

- Components own behavior
- Sequences name messages
- Schemas preserve meaning
- States limit transitions
- Errors trigger actions
- Secrets stop at boundaries
- Latency has a budget
- Tests prove the promise

Detailed design makes independent work safe: people and tools can build against the same promises.

Week 7 Ends with Four Contracts That Agree

By the end of this lecture, students should be able to:

1. Divide CampusConnect into components with one clear responsibility each
2. Model conversations, turns, sources, citations, and fallback states
3. Specify request/response and real-time voice-event contracts
4. Define opening, grounding, citation, abstention, and text-fallback behavior
5. Budget latency and keep secrets out of the browser and repository
6. Turn five questions into repeatable acceptance tests

Lecture Concepts Become Voice-Lab Artifacts

Every concept in this lecture becomes something you can build or verify:

- Component responsibilities → a small CampusConnect architecture sketch
- Interaction sequence → opening, listening, retrieval, answer, citation, fallback
- Data model → Conversation, Turn, Source, Citation, and Feedback
- API and event contracts → example JSON and named failure codes
- Quality attributes → latency, accessibility, security, and recovery rules
- Verification → five questions with expected outcomes

The lab is not a separate activity. It is the smallest implementation of the contracts assembled here.

One User Journey Anchors Every Contract

WHAT THE STUDENT NEEDS

- Ask, “How do I reset my campus password?”
- Hear a concise answer
- See the same answer as text
- Know which guide supports it
- Recover if the microphone fails
- Reach a human when evidence is absent

WHAT CAMPUSCONNECT MUST PROVIDE

- Obtain voice consent
- Capture or accept typed input
- Retrieve approved support content
- Return an answer plus citations
- Abstain when support is insufficient
- Preserve the turn across fallback

The channel may change. The user’s goal, grounding rule, and evidence trail must not.

Components Divide the Voice Journey

EXPERIENCE LAYER

Lovable owns the visible chat, microphone control, transcript, citations, progress states, accessibility, and text fallback.

VOICE + ORCHESTRATION

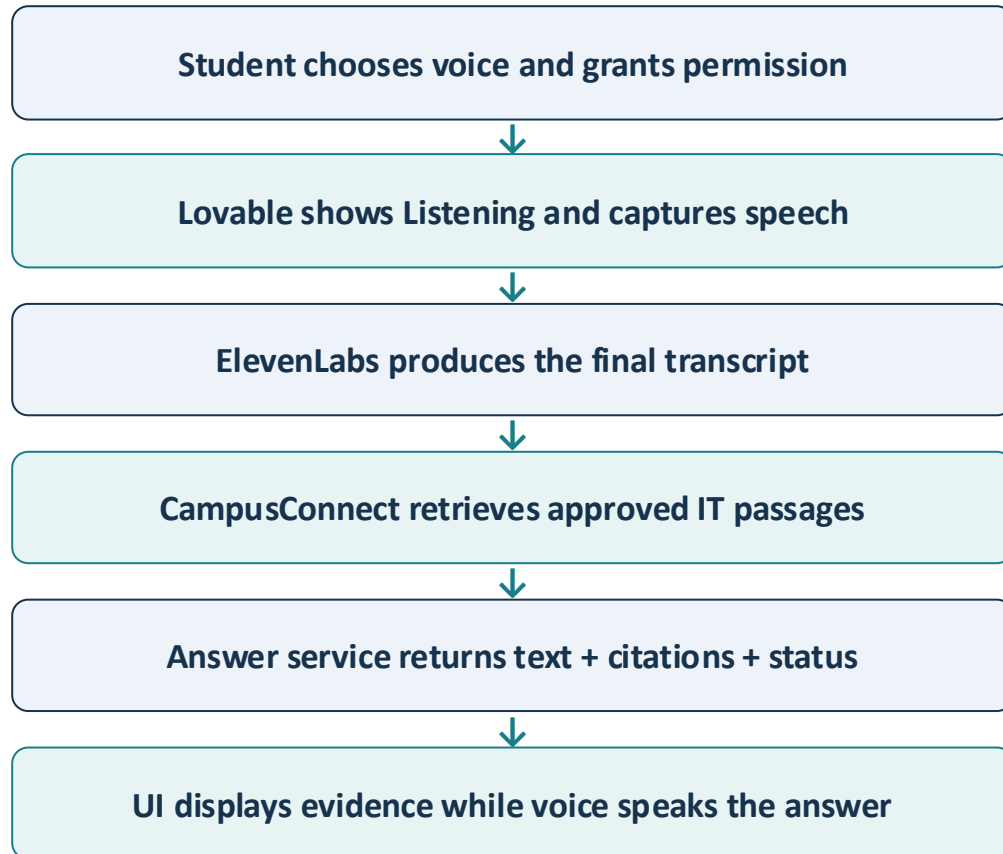
ElevenLabs Agents owns turn-taking, speech recognition, speech output, interruption behavior, and the narrow agent prompt.

KNOWLEDGE + EVIDENCE

CampusConnect retrieval owns approved documents, grounded answers, citation metadata, abstention signals, and test logs.

A component earns a box when it owns behavior, data, or an external boundary—not because a vendor has a logo.

Follow One Grounded Voice Turn through the System



WHAT THE CONTRACT MUST NAME

- A correlation ID for the turn
- Final—not partial—transcript
- Grounded or abstained status
- Source IDs and display labels
- Timeout and retry behavior
- Text fallback without restarting

The AI Periodic Table Supplies a Bounded Route

KNOWLEDGE FOUNDATION

- Approved documents
- Chunking and metadata
- Embeddings and retrieval
- Source identity

INTELLIGENCE ENGINE

- Prompt and context
- Language model
- Structured answer
- Abstention decision

INTERACTION + TRUST

- Voice and text UI
- API and events
- Citations and guardrails
- Evaluation questions

CampusConnect uses only the elements justified by the user journey. The table is a design map—not a feature checklist.

Voice Is a State Machine, Not a Microphone Button



RULES THE TEAM IMPLEMENTS

- Every state is visible in text
- The user can stop speech
- Partial transcript is not submitted
- Timeout leads to an action—not a spinner
- A failed voice turn keeps its transcript
- The next turn may continue in either channel

The Opening Contract Earns Consent and Sets Expectations

IDENTITY + CONSENT

“Hi, I’m the CampusConnect assistant. I answer from approved IT Support guides. May I use your microphone for this conversation?”

CAPABILITY BOUNDARY

The opening says what the agent can do. It does not imply account access, password changes, diagnosis, or authority it does not have.

RECOVERY PROMISE

“You can switch to text at any time.” The text box, transcript, and Stop speaking control remain available from the first turn.

The Grounding Contract Limits What May Be Said

APPROVED CONTEXT

Only institution-approved IT Support documents enter retrieval. Each passage carries a source ID, title, section, and review date.

REQUIRED OUTPUT

The service returns `answer_text`, `citations[]`, `grounded`, `abstained`, and a rationale code—not an attractive paragraph with unknown evidence.

WEAK OR CONFLICTING EVIDENCE

Ask one clarifying question, identify the conflict, or abstain. Never convert model confidence into factual permission to guess.

The Citation Contract Must Work for Ears and Eyes

VOICE CITATION

Speak a short reference: “According to the Password Reset Guide, section 2...” Do not read a long URL aloud.

VISUAL CITATION

Display the full title, section, updated date, and link beside the answer so the student can inspect the evidence.

ALIGNMENT RULE

The spoken answer and displayed answer use the same `answer_text` and citation IDs. A channel may shorten presentation—not change meaning.

Abstention Is a Designed Product Behavior

UNSUPPORTED QUESTION

Example: “Can you tell me another student’s account status?” No approved source and no authorization support that request.

SAFE RESPONSE

“I can’t verify or disclose that. I can help with general account guidance or connect you with IT Support.”

RECORDED NEXT STEP

Return abstained:true, rationale_code:ACCESS_OR_EVIDENCE, no fabricated citation, and a safe escalation option.

Text Fallback Preserves the Journey

Condition	Required interface behavior
VOICE_READY	Microphone works; typed input remains visible.
MIC_DENIED	Explain permission; focus the text box; keep the conversation.
VOICE_TIMEOUT	Preserve the transcript; offer Retry or Continue in text.
TTS_UNAVAILABLE	Display the complete answer and citations; do not replay the question.
RAG_UNAVAILABLE	Abstain; offer official IT Support or human escalation.

Fallback is successful only when the student can finish the same task without repeating the entire conversation.

Accessibility Changes the Interaction Contract

The team converts inclusive intent into observable interface behavior:

- Microphone use is optional; every action works from the keyboard
- A live transcript mirrors both the student and the assistant
- Listening, searching, and error states are announced to assistive technology
- The user can pause or stop speech and request more time
- Citations are meaningful links—not color or icons alone
- Errors appear in both spoken and visible form

Voice is an additional path. It is not an excuse to make the text path second-class.

Data Modeling Starts with Stable Nouns and Lifecycles

The team extracts durable concepts from the CampusConnect voice story:

- Conversation → the continuing user session
- Turn → one question and one resulting outcome
- Source → an approved support document and version
- Citation → the exact source evidence used by a turn
- VoiceSession → channel state, consent, and technical metadata
- Feedback → user rating or correction linked to a turn
- AuditEvent → fallback, error, escalation, or policy decision

Store only what the system must preserve, query, constrain, explain, or improve—and define retention before collecting it.

A Turn Schema Makes One Interaction Concrete

IDENTITY + INPUT

turn_id UUID · conversation_id UUID · channel VOICE | TEXT
question_text · locale · started_at

GROUNDING + RESPONSE

answer_text · citation_ids[] · grounded bool
abstained bool · rationale_code

QUALITY + AUDIT

latency_ms {stt, retrieval, llm, tts, total}
fallback_used · error_code? · user_feedback?

Do Not Let One Schema Pretend to Be Every Schema

Public API / event data	Persistence and model context
Uses client-friendly names and stable enums	Uses normalized IDs, timestamps, and version history
Carries the current turn and citations	Stores conversations, turns, sources, and audit events
Omits private implementation details	May preserve provider, latency, and policy metadata
Validates public input at the boundary	Builds a minimal model context from approved passages
Changes through a versioned contract	Changes through migrations and prompt/evaluation review

Map among the public contract, durable records, and model context deliberately. Do not expose tables—or prompts—as the API by accident.

POST /api/ask Is the Core Text-and-Voice Contract

REQUEST

```
POST /api/ask  
{ conversation_id, text, channel:"VOICE", locale }
```

200 GROUNDED

```
{ turn_id, answer_text, speak_text,  
  citations:[], grounded:true, abstained:false }
```

422 / 503 SAFE FAILURE

```
{ code, message, retryable, fallback_channel,  
  request_id, partial_transcript? }
```

Named Events Make Real-Time Voice Debuggable

CLIENT EVENT

```
{ type:"transcript.final", turn_id,  
  text, captured_at }
```

SERVER EVENT

```
{ type:"answer.grounded", turn_id,  
  answer_text, citations:[] }
```

FAILURE + FALLBACK

```
{ type:"voice.failed", code,  
  retryable, text_fallback:true }
```

Error Codes Must Tell the Interface What to Do

400: CORRECT INPUT

Missing text, unsupported locale, or invalid conversation ID. Show the specific correction and keep the current turn editable.

422: SAFE ABSTENTION

The request is understandable, but evidence or authorization is insufficient. Explain the limit and offer a safe next step.

503: RETRY OR FALL BACK

Voice, retrieval, or model dependency is unavailable. Preserve usable work, mark retryable, and move to text when possible.

A technical error is not complete until the user can understand it and a test can reproduce the expected recovery.

Latency Needs a Budget—Not Hope

A PRACTICAL TURN BUDGET

- Acknowledge within 1 second
- Show retrieval progress immediately
- Target a useful answer within 3 seconds
- Trigger recovery at 8 seconds
- Record each stage separately

THE USER MUST SEE PROGRESS

- Listening → Transcribing → Searching → Answering
- Never show one endless spinner
- Let the user cancel speech
- Keep a partial transcript
- Offer text before patience becomes failure

The numbers are design targets to measure and revise—not promises the team may quietly ignore.

Secrets Stop at the Server Boundary

The team defines where credentials may and may not exist:

- The browser receives no private model or voice API key
- ElevenLabs and other runtime credentials live in Lovable app secrets or a server-side connector
- An Edge Function validates requests before calling private services
- .env.example names variables but contains no values
- GitHub stores code and contracts—not copied secrets
- A leaked key is revoked and rotated immediately; deleting the commit is not enough

A private repository is still not a secret manager.

Fallbacks Preserve the Goal When Dependencies Fail

VOICE UNAVAILABLE

- Keep the transcript
- Continue in text
- Display the same citations
- Offer voice retry later

EVIDENCE UNAVAILABLE

- Do not answer from memory
- Return abstained:true
- Link official support
- Log the failure for review

COMPLETE OUTAGE

- Explain what failed
- Preserve the request ID
- Provide human contact
- Never claim the request succeeded

Graceful degradation means less convenience—not less truth, privacy, or user control.

One Tool Owns Each Lab Responsibility

INTERACTIVE VOICE

ElevenLabs Agents owns speech, turn-taking, interruption, the narrow voice prompt, and the knowledge-connected conversational runtime.

APP SURFACE + INTEGRATION

Lovable owns the accessible web interface, transcript, citation display, progress states, text fallback, and secure connector boundary.

ENGINEERING TRAIL + VIDEO LATER

GitHub stores versioned contracts and test evidence. Google Vids can create a narrated demo or later video artifact—not real-time dialogue.

Adding a tool is justified only when it owns a distinct responsibility that the other tools do not.

Five Questions Turn the Contract into Evidence

Question type	Evidence the team must capture
Known, single source	Correct answer; exact section cited; speech and text agree
Known, multiple sources	Cite both—or state the conflict without blending facts
Ambiguous wording	Ask one useful clarifying question before answering
Unsupported or sensitive	Abstain; no invented citation; provide a safe next step
Voice unavailable	Finish the same journey in text without losing the turn

A demo shows one successful path. These tests ask whether the design remains dependable when the question or channel changes.

Detailed Design Is Done When Others Can Depend on It

“A polished voice demo is interesting. A component, data, interaction, and quality contract that survives failure is engineering.”

- Components own behavior
- Sequences name messages
- Schemas preserve meaning
- APIs and events stay versionable
- Errors trigger actions
- Secrets stay server-side
- Latency becomes measurable
- Access and fallback preserve the goal

Next: build the optional CampusConnect voice path and test five questions against the contract.